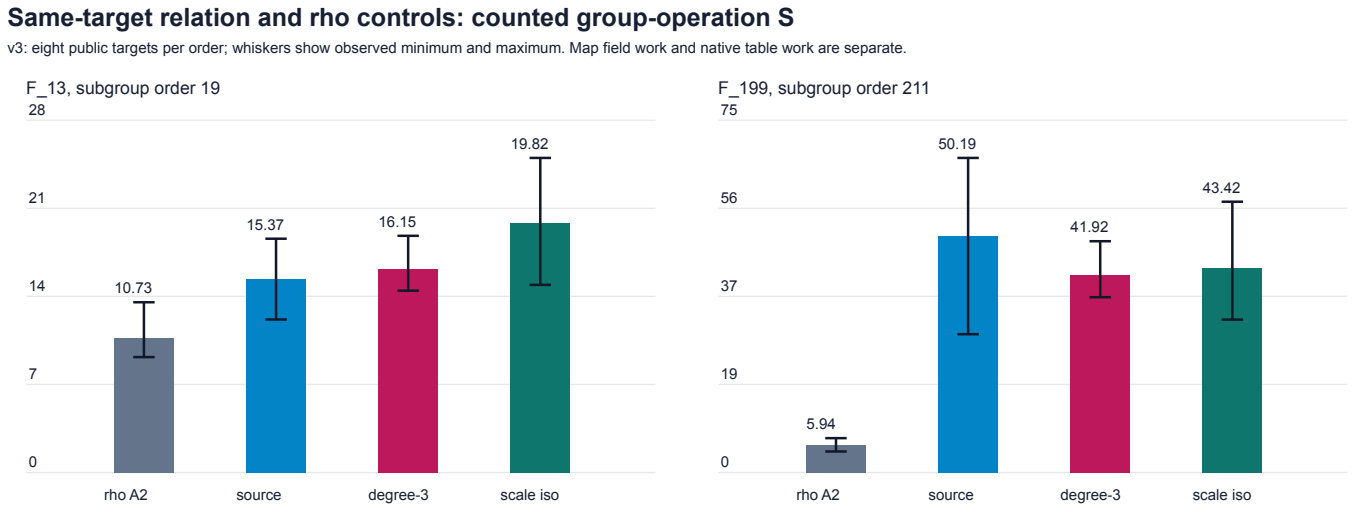
Isomorphism-only transport and six-class rho controls

2026-10-13 local · T-2372 · v3: 80/80 verified source-target answers

The same 16 public source targets were solved through source relations, degree-three transport, scaling-isomorphism transport, negation rho, and six-class rho. The destination pairs have subgroup orders 19 and 211. The isogeny is $\varphi(x,y) = ((x^3+4b)/x^2,\ y(x^3-8b)/x^3)$; the isomorphism is $\psi_u(x,y) = (u^2x, u^3y)$, with u = 2 or 14. The frozen inputs and run parameters are in `PROTOCOL.md`, `REMEDIATION_V2.md`, and `RESTART_V3.md`; the original target set is `../raw-v4.tsv`.

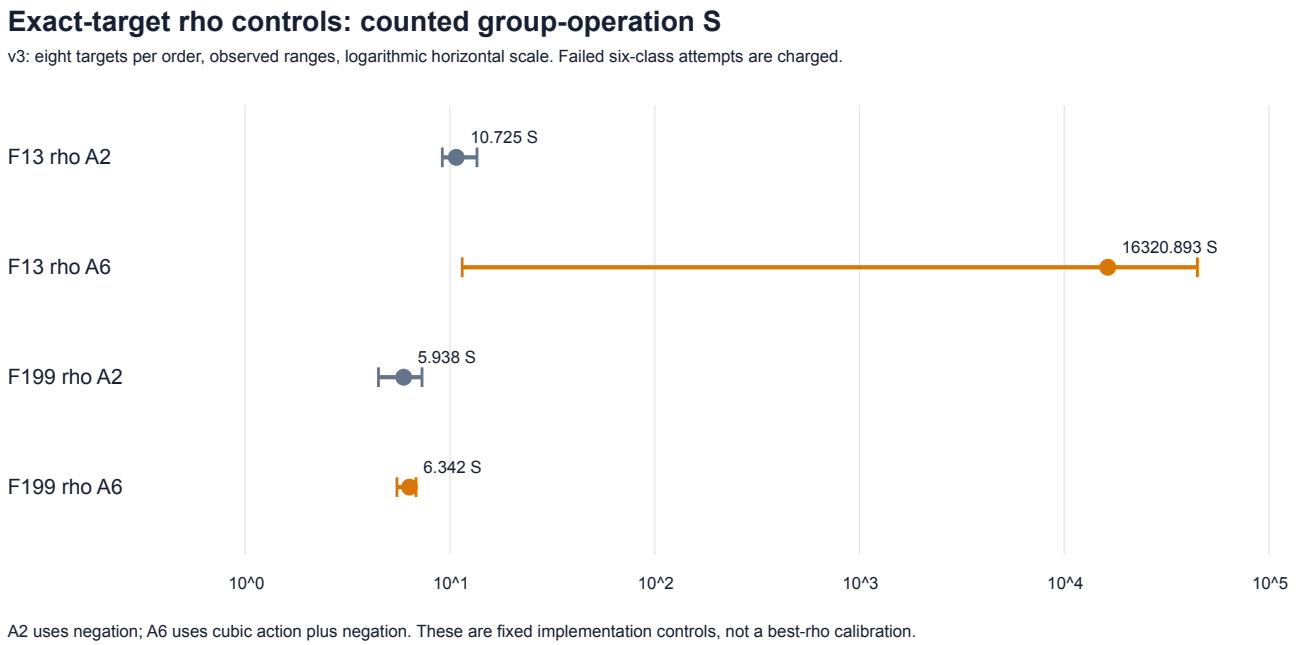
The chart uses S = (group additions + doublings)/√r, including cold relation setup, collection, descent, and verification. Each mean and observed min–max range is over eight targets. The destination arms include source-point cross-verification. Native map, lookup, row, and memory work is recorded separately, so this is a counted-operation diagnostic.



r	arm	mean S	verified
19	rho negation	10.725	8/8
19	rho six classes	16,320.893	8/8
19	source relations	15.371	8/8
19	degree-three relations	16.145	8/8
19	scaling relations	19.816	8/8
211	rho negation	5.938	8/8
211	rho six classes	6.342	8/8
211	source relations	50.186	8/8
211	degree-three relations	41.917	8/8
211	scaling relations	43.423	8/8

Relation collection and charged rho restarts

At order 211, degree-three and scaling-isomorphism relations used 359 and 358 probes over the same eight targets, with 3,102 and 3,107 collection group operations. Their full counted totals were 4,871 and 5,046, so scaling/degree-three = 1.035927 in this unit. At order 19, the totals were 563 and 691, ratio 1.227353. Both maps reach the same destination model, while $\varphi = \psi_u \circ [m]$ on these rational groups ($m = 10 \bmod 19$ and $29 \bmod 211$). The fixed scalar streams therefore need not visit the same destination points. These exact observations do not isolate a degree-three-specific relation effect.

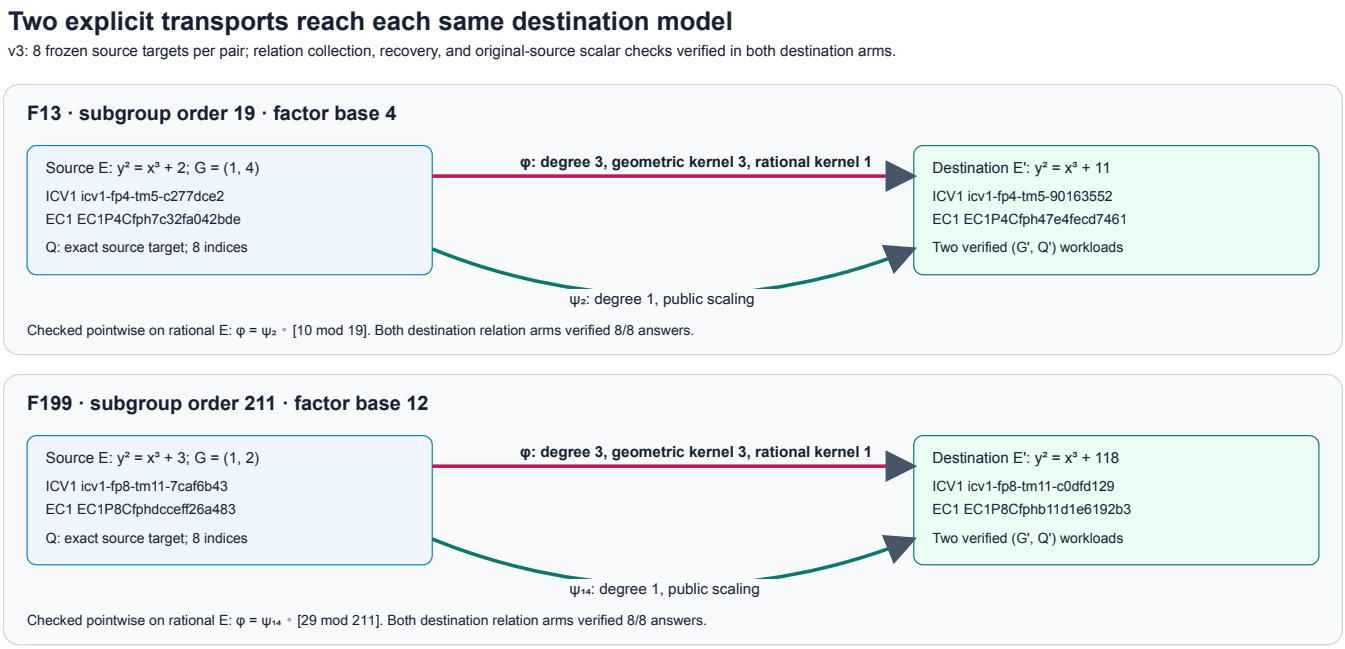


The first two run versions are retained: v1 had 24 verified rows and one six-class cap failure before stopping; v2 had 77 verified rows and three six-class cap failures. V3 followed the precommitted 16-seed restart schedule. Three F13 targets required two attempts; all failed first attempts are charged, giving six-class rho 569,129 counted group operations across eight F13 targets versus 374 for negation rho. At order 211 the respective totals are 737 and 690. These are fixed implementation controls, not a best-rho calibration.

The v3 TSV and log replay byte for byte. An independent Rust checker verified all 80 scalar answers, frozen target coordinates, map images, phase sums, seed/restart accounting, and six-class point multipliers. Its receipt is `independent_v3.log`. Lean checked the conditional coefficient update, scalar-value scaling, and negation lemmas in `SixClassCoefficients.lean`; the actual curve group-action bridge remains open. The host was shared Darwin arm64, L0; no admitted wall-time claim is made. Curve-specific map discovery and exhaustive preflight checks are out of band for these target rows. Full native-work conversion, a base-field non-isomorphic pair, larger orders, independent-host replay, and a multi-point rational torsion fiber remain open. Full evidence and requirement status are in `REPORT.md`.

Explicit transport diagram

The source and destination identities, both map degrees, and the checked relation of the maps on each rational subgroup are shown below. The degree-three kernel has three geometric points and one rational point.



Source: raw-v3.tsv, independent_v3.log, and frozen v4 map/isomorphism checks. Diagram shows named models, not an asymptotic transfer claim.